

6DBSS-ESB and NCC Integration

Safaricom

BSS Transformation

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Revision History

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1.0	23/02/2023		draft

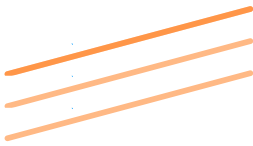
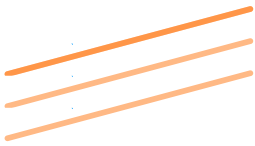


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1 Introduction

1.1 Purpose

The purpose of this Document is to provide the flow for the operations, majorly, between 6DBSS-ESB and NCC.

1.2 Scope

This document shows the Flow for various operations that happens between 6DBSS-ESB and NCC with the Requests.

1.3 Solution Overview

1.3.1 Solution Architecture

1.3.2 Supported Interface

Table 1: Supported interface

Network Element	Protocol	Description
NCC	Rest API	For subscriber profile create/update & provisioning
USSD	Rest API	For subscriber profile create/update & provisioning
TIBCO	Rest API	For subscriber profile create/update & provisioning

1.3.3 Assumptions

The following assumptions have been applied in defining the solution architecture.

ID	Assumptions	Description
BSS-NCC-001	Connectivity between BSS and NCC	Connectivity between BSS-ESB and NCC in all environments like BluePrint, PreProduction, Production is already established

1.3.4 Exclusions

The following exclusions have been applied in the architecture process.

ID	Exclusion	Description
ESB-EX-001		

1.3.5 Architecture and Design Decisions

The following architecture and key design decisions have been made and applied in the development of the architecture.

ID	Decision	Description
ESB-ARC-001		

1.3.6 Architectural Risks & Issues

The following risks and issues have been identified as impacting the architecture.

ID	Risk / Issue	Mitigation Strategy
ESB-RIS-001		

2 Subscriber Association

2.1 6D Subscriber Hierarchy

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2.1.1 6D Prepaid Subscriber

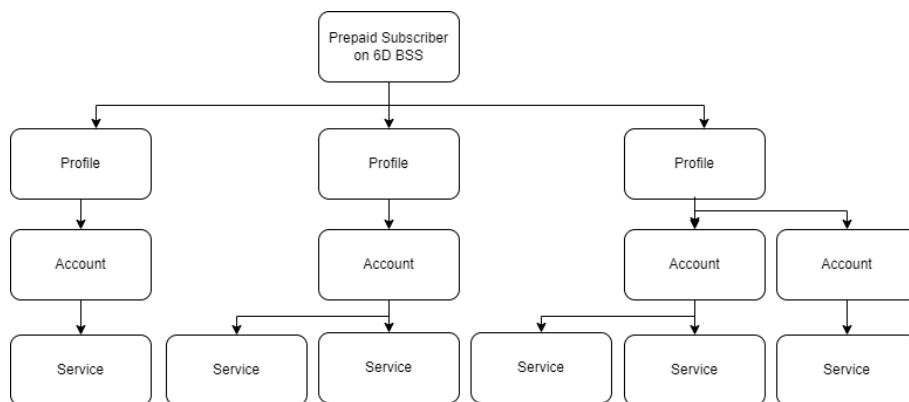


Figure 1: 6D Prepaid Subscriber Hierarchy

2.1.2 6D Postpaid Subscriber

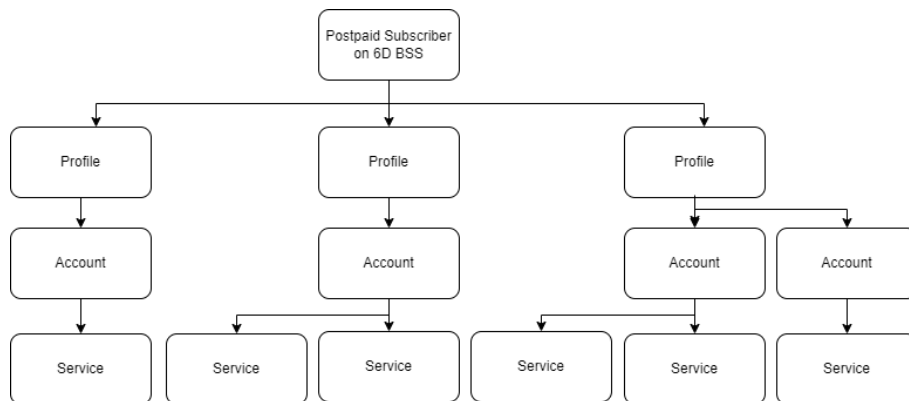


Figure 2: 6D Postpaid Subscriber Hierarchy

2.2 NCC Subscriber Hierarchy

2.2.1 NCC Prepaid Subscriber

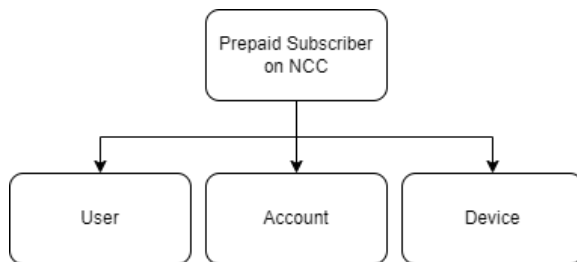


Figure 3: NCC Prepaid Subscriber Hierarchy

2.2.2 NCC Postpaid Subscriber

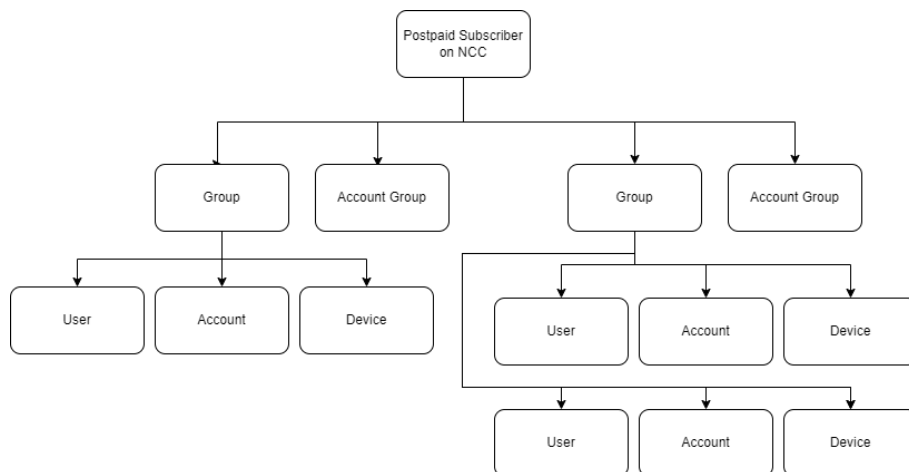


Figure 4: NCC Postpaid Subscriber Hierarchy

3 Prepaid Onboarding

3.1 Prepaid Onboarding Steps

3.1.1 Incoming Req from COM

3.1.2 Request for create an Account

Final Json Request :

```
{
  "id": "A_1041054",
  "dayOfWeek": "SUNDAY",
  "dayOfMonth": "32",
  "accountType": "PRE_P",
  "AID": "AID",
  "customData": {
    "entry": [
      {
        "key": "HYBRID",
        "value": {
          "type": "STRING",
          "value": "FALSE"
        }
      }
    ]
  },
  "hourOfDay": "0",
  "timeZoneId": "Africa/Addis_Ababa",
  "overageLimit": "0",
  "meName": "ig1",
  "lifeCycleNames": {
    "entry": [
      {
        "key": "ENTITY",
        "value": "PRE_ACCOUNT_ELC_V3"
      },
      {
        "key": "PERIOD",
        "value": "PRE_ACCOUNT_PLC_V3"
      }
    ]
  }
} : https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Account
```

3.1.3 Request for create User

Final Json Request :{

```
  "emails": ["safari.com@gmail.com"],
  "accountId": "A_1041054",
  "surname": "Clemens",
  "givenName": "Tatum",
  "meName": "ig1",
  "lifeCycleNames": {
    "entry": [
      {
        "value": "PRE_USER_ELC",
        "key": "ENTITY"
      }
    ]
  },
  "notificationProfile": "PRE_USER_NP",
  "id": "U_1041054",
  "phoneNumbers": ["251700402029"]
} : https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/User
```

3.1.4 Request for create a Device

Final Json Request :

```
{
  "lifeCycleNames": {
    "entry": [
      {
        "key": "ENTITY",
        "value": "PRE_DEVICE_ELC_V3"
      }
    ]
  },
  "syOCSEnabled": true,
  "allowOverage": false,
  "ocsHost": "ig1.mnc02.mcc636.3gppnetwork.org",
  "oc
```

sRealm":"mnc02.mcc636.3gppnetwork.org","identities":[{"identityType":"E164","value":"251700402029"},{"identityType":"IMSI","value":"636024900005352"}],"id":"D_1041054","owner":{"id":"U_1041054"},"customData":{"entry":[{"key":"DATAOOFBFLAG","value":{"type":"STRING","value":"TRUE"}},{"key":"PRIMARYACCOUNTID","value":{"type":"STRING","value":"A_1041054"}}]},"subscriptions":[{"bundle":{"name":"CONSUMER_PREPAY_BASE_PLAN"},"account":{"id":"A_1041054","timeZoneId":"Africa/Addis_Ababa"}],"subscriptionIndex":0.0,"notificationProfile":"PRE_DEVICE_NP","meName":"ig1","language":"ENGLISH": url <https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Device>

3.1.5 Calling getdata Device to get meName

Final url :

[https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Device?processParams={\"id\" : \"D_1041054\", \"detailedQuery\": true }](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Device?processParams={\)

3.1.6 QueryEntityCounterInstance to get usageLimit of SMS COUNTER

Final url :

[https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/EntityCounterInstance?processParams={\"entityId\":\"D_1040720\", \"entityType\":\"DEVICE\", \"counterDefName\":\"SMS COUNTER\"}](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/EntityCounterInstance?processParams={\)

3.1.7 Request for PreActive

Final Json Request :

{\"entityType\":\"com.nokia.tpapps.subscribermodel.model.Device\", \"entityId\":\"D_1041054\", \"triggerEvent\":\"TO_PREACTIVE\", \"lifeCycleType\":\"ENTITY\", \"meName\":\"ig1\"} : url <https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/trigger/LifeCycle>

3.1.8 Request for Active

Final Json Request :

{\"entityType\":\"com.nokia.tpapps.subscribermodel.model.Device\", \"entityId\":\"D_1041054\", \"triggerEvent\":\"TO_ACTIVE\", \"lifeCycleType\":\"ENTITY\", \"meName\":\"ig1\"} : url <https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/trigger/LifeCycle>

3.1.9 QueryEntityCounterInstance to get usageLimit of UNBILLED_USAGE_COUNTER

Final url :

[https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/EntityCounterInstance?processParams={"entityId":"D_1040720","entityType":"DEVICE","counterDefName":"UNBILLED_USAGE_COUNTER"}](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/EntityCounterInstance?processParams={)

3.1.10 Calling getdata Account to get E164

Final url :

[https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Account?processParams={"id":"A_1041054","detailedQuery":true}](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Account?processParams={)

3.1.11 Create Subscription WELCOME_OFFER

Final Json Request :

```
{"accountId":"A_1041054","entities":[{"entityType":"DEVICE","entityId":"D_1041054","subscriptions":[{"bundle":{"name":"WELCOME_OFFER","alternateId":"null:251700402029-null-null-0-Self"}}]}] : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Subscription\_v1
```

3.1.12 Response to COM

3.2 Sample Logs



prepaid_onboarding_
latestlogs_1602.txt

4 Postpaid Onboarding

4.1 Postpaid Onboarding Steps

4.1.1 Incoming Req from COM to NCC

Group Account creation in NCC and Account creation in ERP

4.1.2 Create Profile – group account creation – AG_ and G_

Final Json Request :

```
{
  "account": {
    "dayOfWeek": "SUNDAY",
    "lifeCycleNames": {
      "entry": {
        "key": "ENTITY",
        "value": "POP_GACCOUNT_ELC_U2"
      },
      "key": "PERIOD",
      "value": "POP_GACCOUNT_PLC_U2"
    }
  },
  "id": "AG_ET61501",
  "hourOfDay": "0",
  "dayOfMonth": "1",
  "timeZoneId": "Africa/Addis_Ababa",
  "accountType": "POST_PAID",
  "overageLimit": 0.0,
  "meName": "ig1",
  "group": {
    "lifeCycleNames": {
      "entry": {
        "key": "ENTITY",
        "value": "POP_GROUP_ELC_U2"
      },
      "key": "PERIOD",
      "value": "POP_GROUP_PLC_U2"
    }
  },
  "id": "G_ET61501",
  "name": "G_ET61501",
  "meName": "ig1",
  "subscriptions": {
    "bundle": {
      "name": "GROUP_POSTPAY_BASE_PLAN"
    }
  },
  "notificationProfile": "POP_GROUP_NP"
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Profile
```

4.1.3 ERP Account Creation – TIBCO

Final json request :

```
{
  "accountType": "POST_PAID",
  "contact": {
    "contactMedium": {
      "mediumType": "address",
      "characteristic": {
        "streetName": "Addis Ababa/Kirkos",
        "subCity": "2",
        "poBoxNumber": "",
        "zone": "1",
        "kebele": "Addis Ababa",
        "latitude": "38.7578",
        "houseNumber": "Addis Ababa/Kirkos",
        "woreda": "1",
        "postCode": "Addis Ababa/Kirkos",

```

```

        "primaryPhoneNumber": "251700400374",
        "region": "1",
        "longitude": "8.9806"
    }
  },
  "contactType": "permanentAddress"
},
"name": "Emilio",
"id": "ET61501",
"characteristics": [{
  "name": "class",
  "value": "Individual"
}]
} : url https://tibcobp.blueprint.lab/auth/v1/partyAccount/upsert

```

4.1.4 Response to COM from NCC

Group Account creation in NCC and Account creation in ERP

4.1.5 Incoming Req from COM to NCC

Postpaid User, Account, Device, Subscription creation in NCC

4.1.6 Request for create an Account

Final Json Request :

```

{"id": "A_1041531", "dayOfWeek": "SUNDAY", "dayOfMonth": "1", "accountType": "POST_P
AID", "customData": {"entry": [{"key": "HYBRID", "value": {"type": "STRING", "value": "FALSE"}
}]}, "hourOfDay": "0", "timeZoneId": "Africa/Addis_Ababa", "overageLimit": "100000.0", "m
eName": "ig1", "lifeCycleNames": {"entry": [{"key": "ENTITY", "value": "POP_ACCOUNT_ELC
_U4"}, {"key": "PERIOD", "value": "POP_ACCOUNT_PLC_U4"}]} : https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Ac
count

```

4.1.7 Request for create User

Final Json Request :

```

{
  "emails": ["safaricom@gmail.com"],
  "accountId": "A_1041531",
  "surname": "Ivory",
  "givenName": "Dedrick",
  "meName": "ig1",
  "lifeCycleNames": {"entry": [{

```

```

    "value": "POP_USER_ELC_U2",
    "key": "ENTITY"
  }},
  "notificationProfile": "POP_USER_NP",
  "id": "U_1041531",
  "phoneNumbers": ["251700400278"]
} : https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Us
er

```

4.1.8 Request for create a Device

Final Json Request :

```

{"lifeCycleNames":{"entry":[{"key":"ENTITY","value":"POP_DEVICE_ELC_U2"}],"syOCSEn
abled":true,"allowOverage":true,"ocsHost":"ig1.mnc02.mcc636.3gppnetwork.org","ocs
Realm":"mnc02.mcc636.3gppnetwork.org","identities":[{"identityType":"E164","value":
"251700400278"},{"identityType":"IMSI","value":"636024900005350"}],"id":"D_104153
1","owner":{"id":"U_1041531","customData":{"entry":[{"key":"PRIMARYACCOUNTID","
value":{"type":"STRING","value":"A_1041531"}]},"subscriptions":{"bundle":{"name":"C
ONSUMER_POSTPAY_BASE_PLAN"},"account":{"id":"A_1041531","timeZoneld":"Africa/
Addis_Ababa"}],"subscriptionIndex":0.0,"notificationProfile":"POP_DEVICE_NP","meN
ame":"ig1","language":"ENGLISH"}: url https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/De
vice

```

4.1.9 Counter updates

4.1.9.1 VOICE

Final Json Request for Adding counters:

```

{"id":"VOICECOUNTER","deviceId":"D_1041531": url https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/C
ounter

```

4.1.9.2 DATA

Final Json Request for Adding counters:

```

{"id":"DATACOUNTER","deviceId":"D_1041531": url https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/C
ounter

```

4.1.9.3 SMS

Final Json Request for Adding counters:

```
{ "id": "SMSCOUNTER", "deviceId": "D_1041531": url https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/C
ounter
```

4.1.9.4 UNBILLED_USAGE

Final Json Request for Adding counters:

```
{ "id": "UNBILLED_USAGE_COUNTER", "deviceId": "D_1041531": url https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/C
ounter
```

4.1.9.5 CATEG2COUNTER - OOB Counter

Final Json Request for Adding counters:

```
{ "id": "CATEG2COUNTER", "deviceId": "D_1041531": url https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/C
ounter
```

4.1.10 update Entitycounter instance - category2 overage limit

Final json request :

```
{
  "overageLimit": 100000,
  "entityType": "DEVICE",
  "entityId": "D_1041531",
  "adjustMode": "SET",
  "counterType": "DEVICE",
  "counterDefName": "CATEG2COUNTER"
} : url https://kal-ncc-blp-
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/update/E
ntityCounterInstance
```

4.1.11 deltaUpdate Linking deviceId to the GroupId

Final Json Request : { "id": "G_ET61501", "childDeviceIds": ["D_1041531"]}: url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/deltaUpdate/Group?subAction=Add

4.1.12 Response to COM from NCC

Postpaid User, Account, Device, Subscription creation in NCC

4.1.13 Incoming Req from COM to USSD

CRM Profile Call to USSD

4.1.14 Update CRM Profile Call

Final json request :

```
{ "imsisdn": "251700400278", "groupId": "1", "language": "1", "additionalFields": { "ProfileID": "112452", "OCS_UniqueKey": "1041531", "NetworkType": "1", "BasePlan": "368", "profileType": "0", "P_SubCategory": "23", "P_Category": "231" } } : url  
http://10.32.9.99/apigw/crm/provisioning/v1/Device?processParams={ "id": "D_1040799", "updateSession": "0", "detailedQuery": true }
```

4.1.15 Response to COM from USSD

CRM Profile Call to USSD

4.1.16 Incoming Req from COM to BSS (internal)

Customer activation in BSS

4.1.17 Customer Activation in Bss

Final Json Request :

```
{ "msisdn": "251700400278", "imsi": "636024900005350", "timestamp": "2023-02-24T09:41:27.192Z", "ProfileType": "Consumer", "customerFullName": "Dedrick Ivory", "email": "safaricom@gmail.com", "language": "1", "nationality": "1", "channel": "CRM", "dob": "03-06-1979", "activationType": "Onboarding", "customerStatus": "Active", "customerID": "112452", "reg_document_type": "107", "reg_document_number": "362", "created_user": "APIGW", "document_issue_Date": "2022-03-31T00:00:00.000Z", "isMinor": "No", "IDExpiryDate": "2028-03-31T00:00:00.000Z" }
```

4.1.18 Response to COM from BSS (internal)

Customer activation in BSS

4.2 Sample Logs



5 Connection Migration – Prepaid to Postpaid

5.1 Prepaid to Postpaid Migration Steps

5.1.1 Incoming Req from COM

5.1.2 Calling getdata Account to get meName

Final url:

`https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Account?processParams={"id" : "A_1041111","detailedQuery": true }`

5.1.3 Calling getdata Device to get meName

Final url:

`https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Device?processParams={"id" : "D_1041111","detailedQuery": true }`

5.1.4 Create Profile – group account creation – AG_ and G_

Final Json Request :

```
{
  "account": {
    "dayOfWeek": "SUNDAY",
    "lifeCycleNames": {
      "entry": [
        {
          "key": "ENTITY",
          "value": "POP_GACCOUNT_ELC_U2"
        },
        {
          "key": "PERIOD",
          "value": "POP_GACCOUNT_PLC_U2"
        }
      ]
    },
    "id": "AG_ET61413",
    "hourOfDay": 0,
    "dayOfMonth": 1,
    "timeZoneId": "Africa/Addis_Ababa",
    "accountType": "POST_PAID",
    "overageLimit": 0,
    "group": {
      "lifeCycleNames": {
        "entry": [
          {
            "key": "ENTITY",
            "value": "POP_GROUP_ELC_U2"
          },
          {
            "key": "PERIOD",
            "value": "POP_GROUP_PLC_U2"
          }
        ]
      },
      "id": "G_ET61413",
      "name": "G_ET61413",
      "subscriptions": [
        {
          "bundle": {
            "name": "GROUP_POSTPAY_BASE_PLAN"
          }
        }
      ],
      "notificationProfile": "POP_GROUP_NP",
      "meName": "ig1"
    }
  },
  "url": "https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Profile"
}
```

5.1.5 Changing the Subscriber status to suspend

Final Json Request :

```
{
  "id": "D_1041111",
  "entityType": "Device",
  "patch": [
    {
      "op": "ADD",
      "path": "/customData/entry/3",
      "value": {
        "key": "SUSPENDED",
        "value": {
          "type": "DATE_TIME",
          "value": "16770519958"
        }
      }
    }
  ]
}
```

78"}}}} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch

5.1.6 Session Abort call – device id

Final URL:

https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/sendASRForSubscriber/GySession?processParams={"device":{"id":"D_1041111"}}

5.1.7 Adjusting the balance to Zero

Final json request :

```
{
  "bucket": {
    "@referredType": "Balance",
    "id": "A_1041111"
  },
  "reason": "adjustment customer balance with 0 Euro",
  "amount": {"amount": 0},
  "adjustType": "SET",
  "@type": "adjustBalance",
  "@schemaLocation":
"http://www.nokia.com/ncc/v1.0/TMF_API_Extension_Schema.json#/definitions/adjustBalance",
  "transactionId": "7b9bed80-fff2-465d-954e-e20fb2d93635_1077858968775958528",
  "usageType": "MONETARY"
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/tmf-api/prepayBalanceManagement/v4/adjustBalance
```

5.1.8 Delete Subscription call - Prepaid Base plan – subscription id

Final json request

```
{"entityId":"D_1041111","entityType":"DEVICE","subscriptionIds":["960e99fc-b6d9-4c28-88f1-ddfd033c9be5"]} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/delete/Subscription
```

5.1.9 Patch update at account level - subscriber patch request – account type, day of the month, overage limit

Final Json Request at user level :

```
{
  "id": "A_1041111",
  "entityType": "Account",
  "patch": [
    {
      "op": "replace",
      "path": "/lifeCycleNames",
      "value": {
        "entry": [
          {
            "key": "PERIOD",
            "value": "POP_ACCOUNT_PLC_U4"
          },
          {
            "key": "ENTITY",
            "value": "POP_ACCOUNT_ELC_U4"
          }
        ]
      },
      "op": "replace",
      "path": "/states",
      "value": {
        "entry": [
          {
            "key": "PERIOD",
            "value": {
              "name": "ACTIVE",
              "id": "POP_ACCOUNT_PLC_U4_ACTIVE"
            }
          },
          {
            "key": "ENTITY",
            "value": {
              "name": "ACTIVE",
              "id": "POP_ACCOUNT_ELC_U4_ACTIVE"
            }
          }
        ]
      },
      "op": "replace",
      "path": "/overageLimit",
      "value": "100000.0",
      "op": "replace",
      "path": "/dayOfMonth",
      "value": "1",
      "op": "replace",
      "path": "/accountType",
      "value": "POST_PAID"
    ]
  ]
}
```

url [https://kal-ncc-blp-](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch)

[prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch)

5.1.10 Patch update at User level for notification profile

Final Json Request at user level :

```
{
  "id": "U_1041111",
  "entityType": "User",
  "patch": [
    {
      "op": "replace",
      "path": "/lifeCycleNames",
      "value": {
        "entry": [
          {
            "key": "ENTITY",
            "value": "POP_USER_ELC_U2"
          }
        ]
      },
      "op": "replace",
      "path": "/states",
      "value": {
        "entry": [
          {
            "key": "ENTITY",
            "value": {
              "name": "ACTIVE",
              "id": "POP_USER_ELC_U2_ACTIVE"
            }
          }
        ]
      },
      "op": "replace",
      "path": "/notificationProfile",
      "value": "POP_USER_NP"
    ]
  ]
}
```

url [https://kal-ncc-blp-](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch)

[prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch)

5.1.11 Deltaupdate/deviceModify – allow overage

Final Json Request :

```
{
  "id": "D_1041111",
  "identities": [
    {
      "identityType": "E164",
      "value": "251700402032"
    }
  ],
  "lifeCycleNames": {
    "entry": [
      {
        "key": "ENTITY",
        "value": "POP_DEVICE_ELC_U2"
      }
    ],
    "states": {
      "entry": [
        {
          "key": "ENTITY",
          "value": {
            "barring": false,
            "id": "POP_DEVICE_ELC_U2_ACTIVE",
            "initial": false,
            "name": "ACTIVE",
            "final": false
          }
        }
      ]
    },
    "allowOverage": true,
    "notificationProfile": "POP_DEVICE_NP"
  }
}
```

url [https://kal-ncc-blp-](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/deltaUpdate/Device?subAction=Modify)

[prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/deltaUpdate/Device?subAction=Modify](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/deltaUpdate/Device?subAction=Modify)

5.1.12 Calling getdata Device to get meName

Final url:

[https://kal-ncc-blp-](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Device?processParams={\)

[prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Device?processParams={\"id\" : \"D_1041111\", \"detailedQuery\": true }](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Device?processParams={\)

5.1.13 Changing the Subscriber status to Resume

Final Json Request :

```
{ "id": "D_1041111", "entityType": "Device", "patch": [{"op": "Remove", "path": "/customData/entry/2", "value": {"key": "SUSPENDED", "value": {"type": "DATE_TIME", "value": "1677051998181"}}}] } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch
```

5.1.14 Create subscription- Postpaid base plan Subscription call

Final Json Request for addon:

```
{ "accountId": "A_1041111", "entities": [{"entityType": "DEVICE", "entityId": "D_1041111", "subscriptions": [{"bundle": {"name": "CONSUMER_POSTPAY_BASE_PLAN"}, "alternateId": "TIBCO-BANK:APIGW:7b9bed80-fff2-465d-954e-e20fb2d93635"}]}] } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Subscription_v1
```

5.1.15 Counter updates

5.1.15.1 VOICE

Final Json Request for Adding counters:

```
{ "id": "VOICECOUNTER", "deviceId": "D_1041111" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/Counter
```

5.1.15.2 DATA

Final Json Request for Adding counters:

```
{ "id": "DATACOUNTER", "deviceId": "D_1041111" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/Counter
```

5.1.15.3 SMS

Final Json Request for Adding counters:

```
{ "id": "SMSCOUNTER", "deviceId": "D_1041111" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/Counter
```

5.1.15.4 UNBILLED_USAGE

Final Json Request for Adding counters:

```
{ "id": "UNBILLED_USAGE_COUNTER", "deviceId": "D_1041111" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/Counter
```

5.1.15.5 CATEG2COUNTER - OOB Counter

Final Json Request for Adding counters:

```
{ "id": "CATEG2COUNTER", "deviceId": "D_1041111" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/addDeviceCounter/Counter
```

5.1.16 deltaUpdate Linking deviceId to the GroupId

```
{ "id": "G_ET61413", "childDeviceIds": ["D_1041111"] } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/deltaUpdate/Group?subAction=Add
```

5.1.17 update Entitycounter instance - category2 overage limit

Final json request :

```
{
  "overageLimit": 100000,
  "entityType": "DEVICE",
  "entityId": "D_1041111",
  "adjustMode": "SET",
  "counterType": "DEVICE",
  "counterDefName": "CATEG2COUNTER"
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/update/EntityCounterInstance
```

5.1.18 Respond back to COM

5.2 Sample Logs



CM_Pre2Post_New_Flow.txt

6 Connection Migration – Postpaid to Prepaid (Termination)

6.1 Postpaid to Prepaid Migration Steps

6.1.1 Incoming Req from COM

6.1.2 Calling getdata Device to get meName

Final url:

```
https://kal-ncc-blp-  
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/  
Device?processParams={"id": "D_1041404", "updateSession": "0",  
"detailedQuery":true}
```

6.1.3 QueryEntityCounterInstance to get usageLimit

6.1.3.1 VOICECOUNTER

Final url for QueryEntityCounterInstance to get usageLimit of VOICECOUNTER :

```
https://kal-ncc-blp-  
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/  
EntityCounterInstance?processParams={"entityId":"D_1041404","entityType":"DEVICE"  
,"counterDefName":"VOICECOUNTER"}
```

6.1.3.2 DATACOUNTER

Final url for QueryEntityCounterInstance to get usageLimit of DATACOUNTER :

```
https://kal-ncc-blp-  
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/  
EntityCounterInstance?processParams={"entityId":"D_1041404","entityType":"DEVICE"  
,"counterDefName":"DATACOUNTER"}
```

6.1.3.3 SMS COUNTER

Final url for QueryEntityCounterInstance to get usageLimit of SMS COUNTER :

```
https://kal-ncc-blp-  
prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/  
EntityCounterInstance?processParams={"entityId":"D_1041404","entityType":"DEVICE"  
,"counterDefName":"SMS COUNTER"}
```


6.1.3.4 UNBILLED_USAGE_COUNTER

Final url for QueryEntityCounterInstance to get usageLimit of UNBILLED_USAGE_COUNTER :

https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/EntityCounterInstance?processParams={"entityId":"D_1041404","entityType":"DEVICE","counterDefName":"UNBILLED_USAGE_COUNTER"}

6.1.3.5 CATEG2COUNTER

Final url for QueryEntityCounterInstance to get usageLimit of CATEG2COUNTER :

https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/EntityCounterInstance?processParams={"entityId":"D_1041404","entityType":"DEVICE","counterDefName":"CATEG2COUNTER"}

6.1.4 Calling getdata Account to get meName

Final url:

https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Account?processParams={"id" : "A_1041404","detailedQuery": true }

6.1.5 Adjusting the balance to Zero

Final json request :

```
{
  "bucket": {
    "@referredType": "Balance",
    "id": "A_1041404"
  },
  "reason": "adjustment customer balance with 0 Euro",
  "amount": {"amount": 0},
  "adjustType": "SET",
  "@type": "adjustBalance",
  "@schemaLocation":
"http://www.nokia.com/ncc/v1.0/TMF_API_Extension_Schema.json#/definitions/adjustBalance",
  "transactionId": "CRM53crh8t_1677230611459_1078608390180179968",
  "usageType": "MONETARY"
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/tmf-api/prepayBalanceManagement/v4/adjustBalance
```

6.1.6 LifeCycle Change request - BSS_BARRING

Final Json Request :

```
{ "entityType": "com.nokia.tpapps.subscribermodel.model.Account", "entityId": "A_1041404", "triggerEvent": "BSS_BARRING", "lifeCycleType": "ENTITY", "meName": "ig1" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/trigger/LifeCycle?bridgeEndpoint=true
```

6.1.7 Session Abort call – device id

Final URL:

```
https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/sendASRForSubscriber/GySession?processParams={"device":{"id":"D_1041404"}}
```

6.1.8 deltaUpdate Unlinking deviceid to the GroupId

Final json request :

```
{"id":"G_ET61434","childDeviceIds":["D_1041404"]} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/deltaUpdate/Group?subAction=Remove
```

6.1.9 Deltaupdate/deviceModify – disallow overage and make PRE-Active

Final Json Request :

```
{"id":"D_1041404","identities":[{"identityType":"E164","value":"251700494061"}],"lifeCycleNames":{"entry":[{"key":"ENTITY","value":"PRE_DEVICE_ELC_V3"}],"states":{"entry":[{"key":"ENTITY","value":{"barring":false,"id":"PRE_DEVICE_ELC_V3_PREAMTIVE","initial":false,"name":"PREAMTIVE","final":false}}],"allowOverage":false,"notificationProfile":"PRE_DEVICE_NP"} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/deltaUpdate/Device?subAction=Modify
```

6.1.10 Patch update at account level - subscriber patch request – account type, day of the month, overage limit

Final Json Request at accountLevel :

```
{"id":"A_1041404","entityType":"Account","patch":[{"op":"replace","path":"/lifeCycleNames","value":{"entry":[{"key":"PERIOD","value":"PRE_ACCOUNT_PL_C_V3"}],"key":"ENTITY","value":"PRE_ACCOUNT_ELC_V3"}]},"op":"replace","path":"/states","value":{"entry":
```

```
{
  "key": "PERIOD",
  "value": {
    "name": "INITIAL",
    "id": "PRE_ACCOUNT_PLC_V3_INITIAL"
  },
  "key": "ENTITY",
  "value": {
    "name": "PREACTIVE",
    "id": "PRE_ACCOUNT_ELC_V3_PREACTIVE"
  }
}, {
  "op": "replace",
  "path": "/overageLimit",
  "value": "0"
}, {
  "op": "replace",
  "path": "/dayOfMonth",
  "value": "32"
}, {
  "op": "replace",
  "path": "/accountType",
  "value": "PRE_PAID"
}
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch
```

6.1.11 Patch update at User level for notification profile

Final Json Request at user level :

```
{
  "id": "U_1041404",
  "entityType": "User",
  "patch": [
    {
      "op": "replace",
      "path": "/lifeCycleNames",
      "value": {
        "entry": [
          {
            "key": "ENTITY",
            "value": "PRE_USER_ELC"
          }
        ]
      }
    }, {
      "op": "replace",
      "path": "/states",
      "value": {
        "entry": [
          {
            "key": "ENTITY",
            "value": {
              "name": "Active",
              "id": "PRE_USER_ELC_Active"
            }
          }
        ]
      }
    }, {
      "op": "replace",
      "path": "/notificationProfile",
      "value": "PRE_USER_NP"
    }
  ]
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/patch
```

6.1.12 LifeCycle Change request – TO_ACTIVE

Final Json Request :

```
{
  "entityType": "com.nokia.tpapps.subscribermodel.model.Device",
  "entityId": "D_1041404",
  "triggerEvent": "TO_ACTIVE",
  "lifeCycleType": "ENTITY",
  "meName": "ig1"
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/trigger/LifeCycle?bridgeEndpoint=true
```

6.1.13 Remove Counters

6.1.13.1 VOICE

Final Json Request for removing counters:

```
{
  "id": "VOICECOUNTER",
  "deviceId": "D_1041404"
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/removeDeviceCounter/Counter
```

6.1.13.2 DATA

Final Json Request for removing counters:

```
{
  "id": "DATACOUNTER",
  "deviceId": "D_1041404"
} : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/removeDeviceCounter/Counter
```

6.1.13.3 SMS

Final Json Request for removing counters:

```
{ "id": "SMSCOUNTER", "deviceId": "D_1041404" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/removeDeviceCounter/Counter
```

6.1.13.4 UNBILLED_USAGE

Final Json Request for removing counters:

```
{ "id": "UNBILLED_USAGE_COUNTER", "deviceId": "D_1041404" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/removeDeviceCounter/Counter
```

6.1.13.5 CATEG2COUNTER - OOB Counter

Final Json Request for removing counters:

```
{ "id": "CATEG2COUNTER", "deviceId": "D_1041404" } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/removeDeviceCounter/Counter
```

6.1.14 Create subscription- Prepaid base plan Subscription call

Final Json Request for addon:

```
{ "accountId": "A_1041404", "entities": [ { "entityType": "DEVICE", "entityId": "D_1041404", "subscriptions": [ { "bundle": { "name": "ENTERPRISE_PREPAY_BASE_PLAN", "alternateId": "CRM:b71f022b-caa5-42c0-a2d0-cc59642d2577:CRM53crh8t_1677230611459" } } ] } ] } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/create/Subscription_v1
```

6.1.15 Delete Subscription call - Postpaid Base plan – subscription id

Final json request

```
{ "entityId": "D_1041404", "entityType": "DEVICE", "subscriptionIds": [ "05d5673c-604c-43e5-af44-05d6165fa319" ] } : url https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/delete/Subscription
```

6.1.16 Calling getdata Device to get meName

Final url:

[https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Account?processParams={"id": "A_1041404", "detailedQuery": true }](https://kal-ncc-blp-prov.mnc02.mcc636.3gppnetwork.org/services/ServiceManager/subscriber/getData/Account?processParams={)

6.1.17 Respond back to COM

6.2 Sample Logs



7 References and Definitions

7.1 Referenced Documents

Ref	Document Title	Version
1		{attach document}

7.2 Glossary

Term, Abbreviation or Acronym	Definition
BSS	Business Support System
NCC	Nokia Converged Charging